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Web Development Project

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1. Abstraction / Introduction

This project is a web application developed for a Private Campus to manage student information. It includes user authentication, student registration, searching, updating, and deleting student data. The website is designed with a unique interface and easy-to-use features. The application is built using HTML, CSS, PHP, MySQL and hosted using WAMP server.

2. Objective of the Project

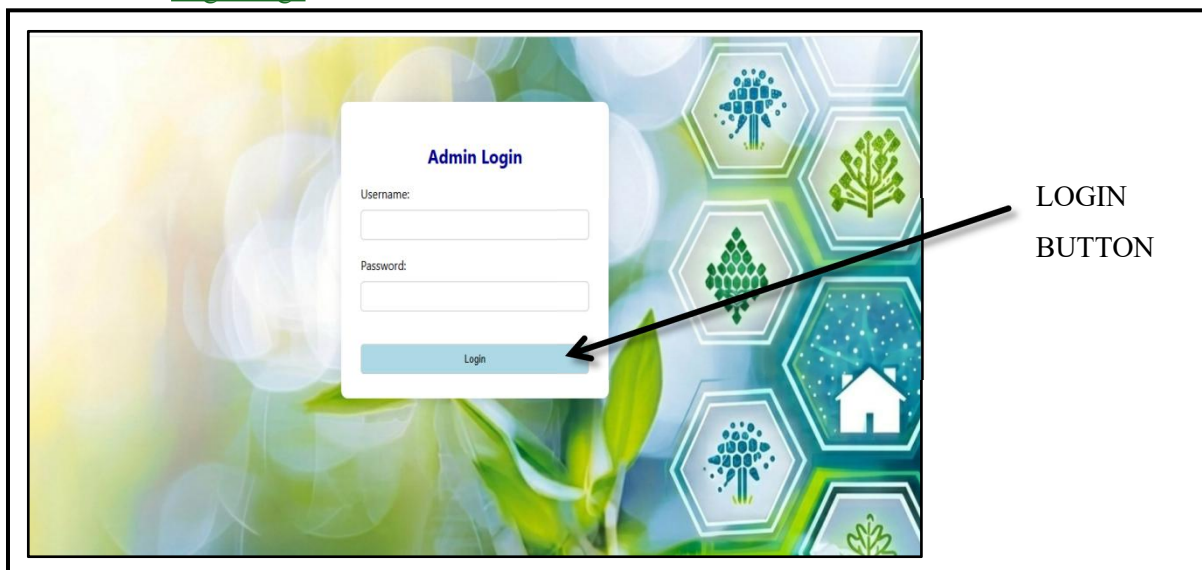
- To design and develop a student management website for a private campus.
- To allow admin to register, view, update, and delete student records easily.
- To use a simple and user-friendly interface.
- To store all student data securely in a database.
- To ensuring security through an access system so that unauthorized persons cannot access data.

3. Tools and Technologies Used

Technology	Description
HTML/CSS	Frontend design and layout
PHP	Server-side Scripting Language
MySQL	Database to store student records

4. Project UI and Explanation

a. Login Page



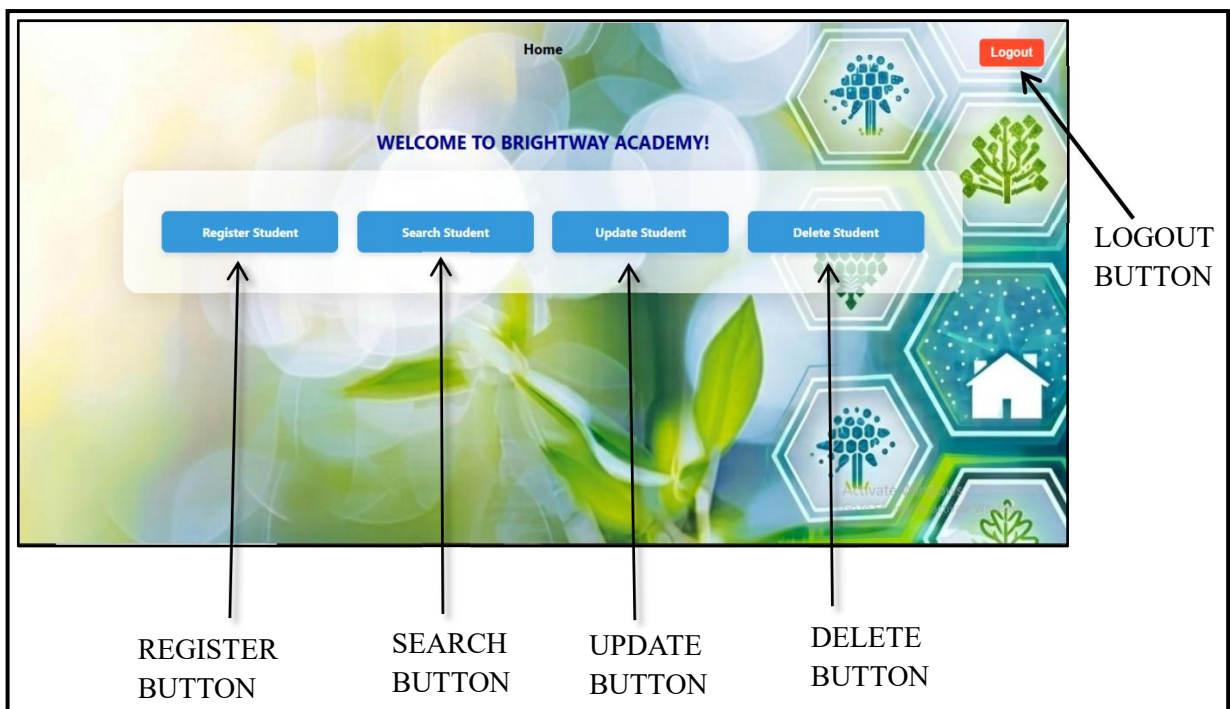
1 LogIn Page

- The Login Page is the first page of the website where the **admin** (or user) must enter a **Username** and **Password** to access the system.

How Does It Work

1. The admin opens the website and sees the login form.
2. The form has two main input fields:
 - ✧ **Username** field - *for entering the admin username.*
 - ✧ **Password** field - *for entering the admin password.*
3. After entering the details, the admin clicks the Login button.
 - ✧ If the username and password are correct (checked using PHP and the database), the admin is redirected to the dashboard.
 - ✧ If the details are incorrect, an error message like "Invalid Login" is shown.

b. Dashboard



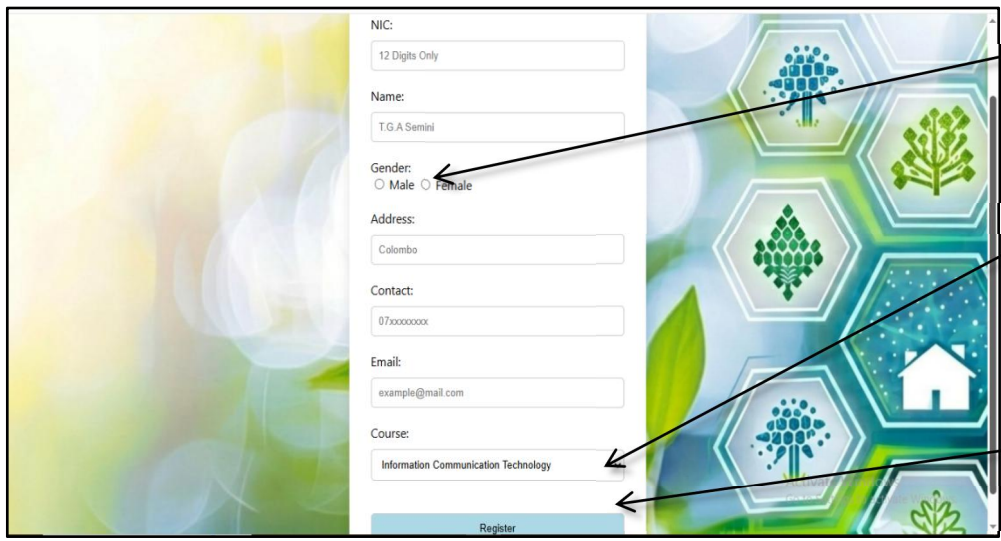
2 Dashboard

- The dashboard features a centralized student management system that allows administrators to **register**, **search**, **update**, and **delete** student records from a single, intuitive interface. This 'all-in-one' approach eliminates the need for complex navigation, making administrative tasks faster and more accurate.

Key Features :

- ◆ Registration: *Quickly add new students to the system.*
- ◆ Search: *Instantly find any student record using ID.*
- ◆ Update: *Edit student profiles and information on the fly.*
- ◆ Delete: *Remove student records with a single click.*
- ◆ Logout button: *For logout from the system.*
- ◆ Home button: *For enter the dashboard easily.*

c. Student Registration Page



The screenshot shows a registration form on the left and a decorative sidebar on the right. The form includes fields for NIC (12 Digits Only), Name (T.G.A Semini), Gender (radio buttons for Male and Female), Address (Colombo), Contact (07xxxxxxxx), Email (example@mail.com), and Course (Information Communication Technology). A 'Register' button is at the bottom. The sidebar on the right has a vertical list of icons: a group of people, a tree, a house, and another group of people. Annotations with arrows point to the 'Female' radio button, the sidebar icons (labeled 'DROP DOWN MENU'), and the 'Register' button (labeled 'REGISTER BUTTON').

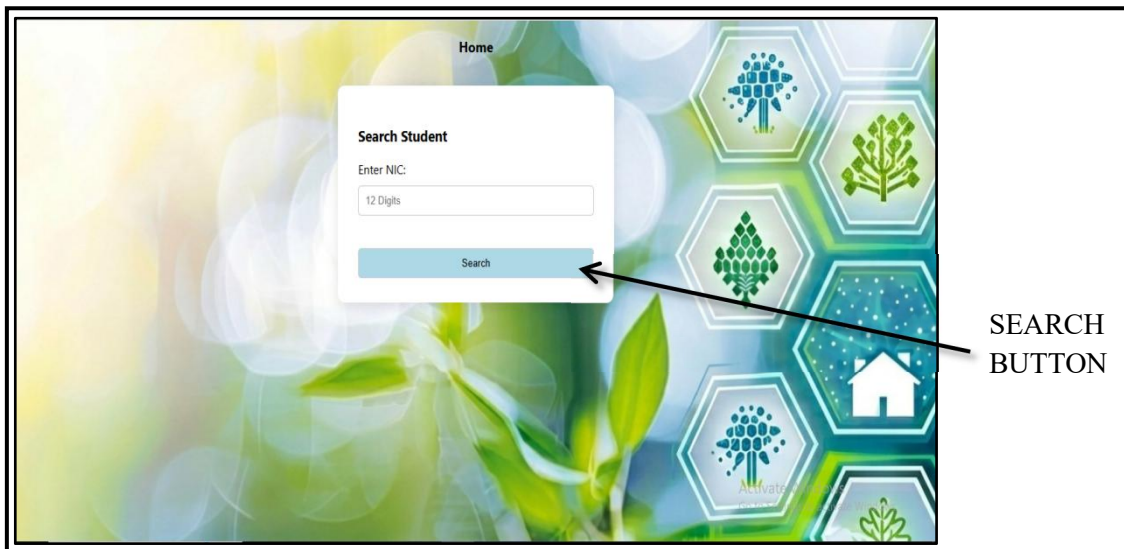
3 Register page

- The **Registration Page** is the interface where admins can add new students to the system by entering their essential details.

How Does It Works

1. The admin enter the dashboard and see the student register button. Then go to Register form.
2. The form has some input fields:
 - ✧ **NIC** field - *For entering the student NIC number.*
 - ✧ **Name** field - *For entering the student name.*
 - ✧ **Gender** field - *For selecting the student gender withing **radio button**.*
 - ✧ **Address** field - *For entering the student address.*
 - ✧ **Contact** field - *For entering the student contact number.*
 - ✧ **Email** field - *For entering the student email info.*
 - ✧ **Course** field - *For selecting the course who prefer to enroll withing **drop down menu**.*
- If the field not fill, it shows “**Please fill out the field**”.
3. After entering the details, the admin clicks the **Register** button.
 - If the NIC number same, an error message like “**Error**” is shown.
 - If enter the correct details of student entered to the database them.

d. Search Student Page



Home

Search Student

Enter NIC:

12 Digits

Search

SEARCH
BUTTON

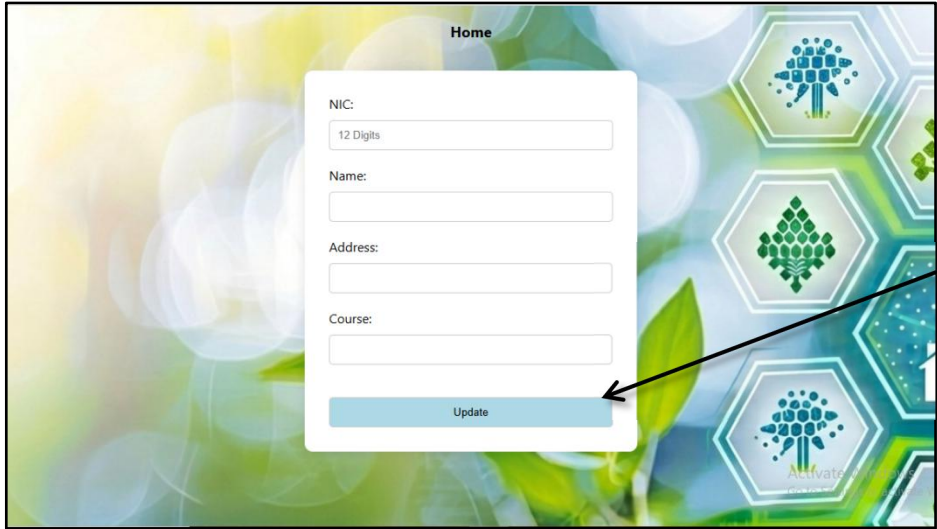
4 Search page

- The Search **Student Page** allows the admin to quickly find specific student records by entering a unique identifier like a **Student NIC**

How Does It Works

1. The admin can opens the search form.
2. It has one field:
 - ✧ **NIC** field: For entering the student NIC number and get the details.
3. After entering the NIC, the admin clicks the **Search** button.
 - If the NIC is correct (checked using PHP and the database), the admin can see the details of the student.
 - If the NIC number incorrect, an error message like “No Student Found with this NIC” is shown.

e. Student Update Page



The screenshot shows a web application interface for updating student information. At the top, there is a 'Home' link. Below it, a form is displayed with the following fields: 'NIC:' with a placeholder '12 Digits', 'Name:', 'Address:', and 'Course:'. Each field has a corresponding input box. At the bottom of the form is a blue 'Update' button. An arrow points from the text 'UPDATE BUTTON' to this button. The background of the page features a green and blue abstract design with hexagonal patterns and a tree icon. The text 'Activats' is visible in the bottom right corner of the page.

5 Update page

- The **Update Page** is the section of the website where the admin can modify or change existing student information in the system.

How Does It Works

1. The admin opens the update form.
2. The form has four input fields:

- ✧ **NIC** - For edit the student NIC number.
- ✧ **Name** - For edit the student name.
- ✧ **Address** - For edit the student address.
- ✧ **Course** - For edit the student course.

3. After entering the correct details, the admin clicks the **Update** button.

- If the details are correct, and update the database.

f. Student Delete Page



6 Delete Page

- The **Delete Page** is the section of the website where the admin can remove old or incorrect student records from the system.

How Does It Works

1. The admin opens the delete form.
2. The form has one main input field:

- ✧ **NIC** - For entering the student NIC number to delete records.

3. After entering the delete button, database will be remove the details about that NIC number.

- **Data Removal:** Permanently deletes a student's profile from the database.

- **Database Maintenance:** Helps to keep the system clean by removing unnecessary information.
- **Control:** Ensures that only authorized admins can remove records to prevent data loss.

5 Database Design

Table Name	Fields
students	<u>Nic</u> , name, gender, address, contract, email, course
users	Id, username, password

6 Challenges Faced and Solution

Challenge	Solution
Wamp Server installation error	Watch the YouTube video and fix
PHP-MySQL connection error	Carefully checked connection code and database settings
Link the background image	Get help from W3 school
Style.css trouble	Get instructions from Chatgpt
Add a student icon to the search result card trouble	Checked the php codes help with Chatgpt

7 Submit Web Application Development Project

Project File	https://drive.google.com/drive/folders/1r3ErADwalavRZvXLRzsi3lqOt0vPN4vJ?usp=drive_link
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